

Shorful Alam

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RESEARCH FOCUS

Bayesian statistics, empirical Bayes methods, and hierarchical/small-area estimation, with applied experience developing INLA-based models for sparse demographic and health survey data. Current work focuses on adult mortality estimation from DHS sibling survival histories and the comparison of exchangeable and age-structured Bayesian priors.

Research Interests: Bayesian statistics · empirical Bayes · hierarchical models · Bayesian computation (INLA, Stan) · small-area estimation · survival analysis · statistical methodology for sparse and complex data

PUBLICATIONS

Alam, S., Haque, N., Mazed, A., Ahmed, M. F. “*Exchangeable versus Age-Structured Priors in Small-Area Estimation of Adult Mortality from Sibling Survival Histories.*” (**First and Corresponding Author**) Preprint: [SocArXiv](#) · Under review at *Demography*

Alam, S., Haque, N., Mazed, A., Mayen, M. S. A., Kowser, M. A., Hassan, M. “*Network-Adjusted Empirical Bayes Estimation of Age-Specific Adult Mortality from DHS Sibling Survival Histories.*” (**First and Corresponding Author**) Preprint: [Research Square](#) · Under review at *Population Health Metrics*

Alam, A.T.M.S., **Alam, S.**, Mobasshira, K., Hasan, M.N., Chowdhury, M.A.B., Uddin, M.J. “*Decomposing Urban-Rural Inequalities in the Utilization of Maternal Health Care Services in Bangladesh.*” Published · [Heliyon](#)

Mazed, A.*, **Alam, S.***, Ali, R., Uddin, M.J. “*Health Literacy and Associated Factors among University Students in Bangladesh.*” (*Co-first authorship) Forthcoming · *BMC Public Health*

Mazed, A.*, **Alam, S.***, Ali, R., Ahmed, N.U., Uddin, M.J. “*Health Literacy Level among University Students: A Systematic Review.*” (*Co-first authorship) Under review

Noor, S.T.A., Islam, R.B., Mostafa, M.S., **Alam, S.**, Islam, A.A. “*Spatial Inequalities in Barriers to Healthcare Utilization Among Women in Bangladesh: A Nationwide Analysis.*” Preprint: [Research Square](#) · Under review

Mayen, M.S.A., Mazed, A., Chowdhury, A., **Alam, S.**, Chowdhury, A.B., Uddin, J. “*Examining the Prevalence and Determinants of Treatment-Seeking for ARI, Fever, and Diarrhea in Children Under Five in Bangladesh: A Pooled Analysis of DHS Data (2011–2022).*” Under review

CONFERENCE & POSTER PRESENTATIONS

Treatment-Seeking Behaviors for ARI, Fever, and Diarrhea in Children Under Five in Bangladesh
8th International Conference on The Role of Statistics and Data Science in 4IR · 2025

Urban-Rural Inequalities in the Utilization of Maternal Health Care Services Across Bangladesh (Poster)
International Conference of Physical Science, SUST · 2022

Risk Factors of COVID-19 Pandemic's Footprints in Bangladesh: An Application of Negative Binomial Regression
International Seminar & Stat-Fest, SUST · 2023

RESEARCH EXPERIENCE

Research Assistant — Health Literacy and Associated Factors among University Students in Bangladesh
SUST Research Center · Supervisor: Dr. Md Jamal Uddin · Jul 2022 – Jun 2023

- Co-developed the research proposal and conducted a systematic review.
- Analyzed survey data and contributed to two manuscripts and multiple conference presentations.
- Supervised junior students contributing to the project.

Research Group Member — BEpPh Research Team
Faculty-led research group, Bangladesh · Jan 2022 – Present

- Collaborate on applied health research projects in biostatistics and epidemiology.
- Contribute to data analysis, literature reviews, and manuscript preparation.

EDUCATION

M.Sc. in Statistics CGPA 3.46 / 4.0
Shahjalal University of Science and Technology, Sylhet, Bangladesh 2023 – 2024

B.Sc. in Statistics CGPA 3.40 / 4.0
Shahjalal University of Science and Technology, Sylhet, Bangladesh 2018 – 2023

EMPLOYMENT

Assistant Teacher (Government Employee)
Ministry of Primary and Mass Education, Government of Bangladesh · Mar 2024 – Present

- Serve as class teacher, delivering curriculum-based lessons and maintaining classroom records.
- Monitor student progress, provide mentorship, and communicate with parents as needed.

TECHNICAL SKILLS

Bayesian & Statistical Methods: Hierarchical Bayesian modeling (R-INLA: IID and RW2 priors, penalized-complexity priors), empirical Bayes methods (SURE-based heteroscedastic shrinkage), small-area estimation (Fay–Herriot models), survival analysis, Monte Carlo simulation, design-based bootstrap variance estimation, regression, time-series analysis.

Programming & Software: R (*INLA, tidyverse, ggplot2, caret, surveybootstrap*), Python (*NumPy, pandas, scikit-learn, TensorFlow/Keras, Matplotlib, Seaborn*), SQL, Stata, SPSS, LaTeX.

Applied Research Areas: Demography, public health, health survey data, spatial and inequality analysis.

Machine Learning: Supervised and unsupervised learning (Random Forest, XGBoost, SVM, k-NN, clustering, PCA).

ACTIVITIES

Head, Sound Management Committee, International Seminar - 2023, SUST. **Volunteer, KIN (2019–2022)** — fundraising and community education. **Departmental Football Team** (2019–2023), Co-Captain.

REFERENCES

Dr. Md Jamal Uddin
Professor, Department of Statistics, SUST, Sylhet · jamal-sta@sust.edu · Website: [Dr. Md Jamal Uddin](#)

Dr. S. M. Khurshid Alam
Professor, Department of Statistics, SUST, Sylhet · khurshid@sust.edu · Website: [Dr. S. M. Khurshid Alam](#)